

6 (a) Thermoregulation is the control of the core temperature of the body.

Thermoreceptors send information about a change in core temperature to the hypothalamus. The hypothalamus controls the core temperature by sending impulses to activate several physiological responses.

Table 6.1 lists physiological responses to a **decrease** in core temperature.

Complete Table 6.1 to explain how each response contributes to maintaining a core temperature within narrow limits.

Table 6.1

response	explanation
blood vessels near the surface of the skin constrict (narrow)	
shivering occurs	
hairs rise	
reduction in the production of sweat	

[4]

(b) Animals can also use behaviour to adapt to temperature changes.

Suggest why some animals curl up their bodies in cold weather.

.....
..... [1]

- (c) The Sitatunga deer, *Tragelaphus spekii*, and the Nile monitor lizard, *Varanus niloticus*, live in Africa.

Fig. 6.1 shows a Sitatunga and Fig. 6.2 shows a Nile monitor.



Fig. 6.1



Fig. 6.2

An investigation was carried out to measure the core temperature of the Sitatunga and the Nile monitor as the environmental temperature was increased.

The results are shown in Fig. 6.3.

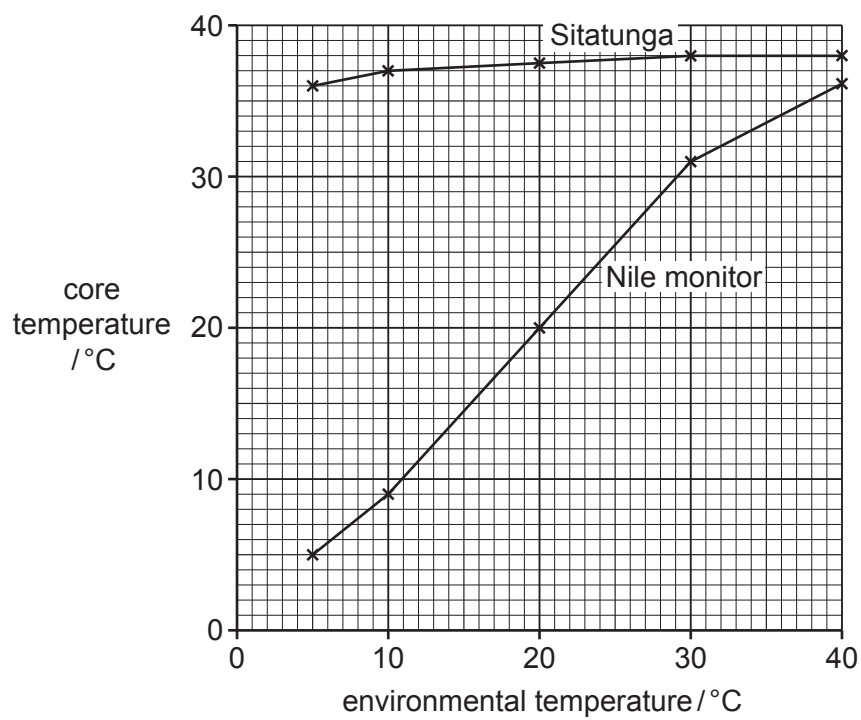


Fig. 6.3

(i) Describe the results shown in Fig. 6.3.

.....

.....

.....

.....

.....

.....

.....

..... [3]

(ii) When the environmental temperature is low, Nile monitors are relatively inactive.

Suggest **one** disadvantage to the Nile monitor of being inactive.

.....

.....

..... [1]

(d) Thermoregulation relies on negative feedback.

Explain what is meant by negative feedback.

.....

.....

.....

.....

.....

.....

..... [3]

[Total: 12]